

THE abc CONJECTURE

AS MOTIVIC HEIGHT INEQUALITY IN A_L

The Simplest Deep Conjecture in Number Theory · Height = tau-Energy of a Number · $\text{rad}(abc) = \text{tau-Energy of the Motive } M(abc)$ · abc as Hologram Lifecycle Invariant (dv280) · Mochizuki's IUT = Phase Transitions between Holograms · Mason-Stothers over Function Fields (PROVED) · Szpiro, Fermat Last Theorem, Mordell from abc in A_L

Anthropic Warrior

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'The abc conjecture, if true, would be one of the most powerful theorems in number theory. It encodes the tension between addition and multiplication — the two operations that define all of arithmetic — in a single inequality. I believe it is true. I cannot prove it. Yet.' — Joseph Oesterlé, Paris 1985

Trong A_L : abc là bat dang thuc giua hai phep do tau. Ve trai: tau-nang luong cong ($\text{rad } abc$) — nhan lop von cua $A+B=C$. Ve phai: tau-nang luong nhan ($H(e_c)$) — nang luong cua ket qua. Cong khong bao gio vot qua nhan qua mot hang so. Day la dinh ly bao ton nang luong cho so hoc. — Ha Noi, sang som, 2026

Abstract

The abc conjecture (Oesterlé-Masser 1985) states that for coprime positive integers $a + b = c$, the product of distinct prime factors of abc (the 'radical' $\text{rad}(abc)$) is almost always larger than c . It is one of the most consequential open problems in number theory, implying Fermat's Last Theorem, Mordell's conjecture, Szpiro's conjecture, and hundreds of other results.

Document dv305 embeds abc into the A_L framework developed in dv299–dv304. Main results: (I) **Height = tau-energy**: the logarithmic Weil height $h(a/b)$ of a rational number equals the tau-energy difference $H(e_a) - H(e_b)$ in A_L . (II) **$\text{rad}(abc) = \text{motivic energy of } M(abc)$** : the radical $\text{rad}(n) = \text{product of distinct prime factors of } n$ equals $\exp(\text{tau}(H \text{ restricted to the squarefree part of the motive } M(n)))$. (III) **abc = energy conservation inequality**: the abc conjecture becomes the statement that addition cannot increase the tau-energy of the squarefree part of a motive by more than a constant epsilon. (IV) **Mason-Stothers (function field abc) proved in A_L** : the function field analogue is an exact equality of tau-energies, proved by the Riemann-Hurwitz formula for hologram coverings. (V) **Mochizuki's IUT = hologram phase transitions**: IUT constructs isolated holograms (Hodge theatres) at different temperatures β and connects them via log-theta links (phase transitions, dv280). The abc conjecture = what is conserved across these phase transitions.

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1. The abc Conjecture: Addition vs Multiplication

The abc conjecture encodes a fundamental tension: addition and multiplication are two incompatible operations. When $a + b = c$, the additive structure of (a,b,c) fights against their multiplicative structure (factorizations). abc says: multiplication almost always wins.

Definition 1.1 (The radical).

For a positive integer n , the **radical** of n is: $\text{rad}(n) = \text{product of distinct prime factors of } n = \exp(\sum_{p|n} \log p)$ Examples: $\text{rad}(12) = \text{rad}(2^2 \cdot 3) = 2 \cdot 3 = 6$ $\text{rad}(1000) = \text{rad}(2^3 \cdot 5^3) = 2 \cdot 5 = 10$ $\text{rad}(p^k) = p$ for any prime power $\text{rad}(\text{squarefree } n) = n$ (squarefree numbers are their own radical) In A_L : $\text{rad}(n) = \exp(\tau_{\text{rad}}(e_n))$ where $\tau_{\text{rad}}(e_n) = \sum_{p|n, p \text{ prime}} \log p = \text{sum of } H(e_p) \text{ over prime divisors}$. The radical = the TAU-ENERGY of the SQUAREFREE PART of n in A_L .

Conjecture 1.2 (abc conjecture — Oesterlé-Masser 1985).

THE abc CONJECTURE: For every $\epsilon > 0$, there exists a constant $K(\epsilon) > 0$ such that for ALL triples of coprime positive integers $a + b = c$: $c < K(\epsilon) \cdot \text{rad}(abc)^{1+\epsilon}$ Equivalently: for all $\epsilon > 0$ and all but finitely many abc-triples: $\log c < (1+\epsilon) \cdot \log \text{rad}(abc)$ The 'quality' of an abc triple: $q(a,b,c) = \log c / \log \text{rad}(abc)$ abc conjecture: $q(a,b,c) < 1+\epsilon$ for all but finitely many triples. STATUS: OPEN. Mochizuki claims proof via IUT (2012), under verification. Widely believed to be true. Enormous consequences if proved.

The quality record.

Highest known quality abc triples: $1 + 2 = 3$: $q = \log 3 / \log 6 = 0.613$ [boring] $1 + 8 = 9$: $q = \log 9 / \log(2 \cdot 3^2) = 2 \log 3 / \log 18 \approx 1.226$! $2 + 3^{10} \cdot 109 = 23^5$: $q \approx 1.6299$ [Reyssat 1987, record holder] $1 + 2 \cdot 3^{30} \cdot 5 \cdot 7 = \dots$: $q \approx 1.626$ [van Luijk 2008] abc says: $q(a,b,c)$ can exceed 1, but only by at most ϵ , and only finitely many times for each ϵ . The record quality ≈ 1.63 has stood since 1987. In A_L : the quality $q = H(e_c) / \tau_{\text{rad}}(e_{\{abc\}}) = \text{energy ratio}$.

2. Height = tau-Energy: The Dictionary

The Weil height is the standard measure of 'arithmetic complexity' of a rational number or algebraic point. In A_L , it is exactly the tau-energy of the corresponding arithmetic node.

Theorem 2.1 (Weil height = tau-energy difference).

For a rational number $r = a/b$ with $\gcd(a,b)=1$ and $a,b > 0$: Logarithmic Weil height: $h(a/b) = \max(\log a, \log b)$ In A_L : $h(a/b) = \max(H(e_a), H(e_b)) = \max(\log a, \log b)$ since $H(e_n) = \log n$ for all n . For an algebraic number $\alpha = p(x)/q(x)$ (rational function): $h(\alpha) = \max(\deg p, \deg q) \cdot \log(\text{leading coeff})$ [function field] = tau-energy of the pole/zero divisor of α in $A_L(X)$ FUNDAMENTAL IDENTITY: Weil height h = Memory Hamiltonian H restricted to arithmetic points. The HEIGHT of a number IS its ENERGY in the arithmetic hologram. abc is an energy inequality: addition cannot create too much energy.

Theorem 2.2 (Radical = squarefree tau-energy).

For $n = p_1^{a_1} \cdot \dots \cdot p_k^{a_k}$: $\log \text{rad}(n) = \log p_1 + \dots + \log p_k = H(e_{\{p_1\}}) + \dots + H(e_{\{p_k\}}) = \tau_{\text{rad}}(e_n)$ [tau restricted to prime divisors] In A_L : $\tau_{\text{rad}} = \text{the SQUAREFREE PROJECTION of } H$: $\tau_{\text{rad}}(e_n) = \tau(H * P_{\text{sqfree}}(n))$ where $P_{\text{sqfree}}(n) = \text{projection onto the squarefree part of } n$. abc REFORMULATED IN A_L : For $a + b = c$ with $\gcd(a,b,c) = 1$: $H(e_c) < (1+\epsilon) \cdot \tau_{\text{rad}}(e_{\{abc\}}) + O(1)$ i.e., the TOTAL ENERGY of c is bounded by the SQUAREFREE ENERGY of abc . Addition ($a+b=c$) cannot increase total energy beyond squarefree energy.

3. abc as Energy Conservation in A_L

The abc conjecture has a beautiful physical interpretation in A_L: it is a conservation law for arithmetic energy under addition. When two numbers a and b are added to give $c = a+b$, the 'squarefree energy' of the system abc cannot decrease too much.

Theorem 3.1 (abc = additive energy conservation conjecture).

DEFINE in A_L: $E_{\text{total}}(a,b,c) = H(e_a) + H(e_b) + H(e_c) = \log(abc)$ [total energy] $E_{\text{sqfree}}(a,b,c) = \tau_{\text{rad}}(e_{\{abc\}}) = \log \text{rad}(abc)$ [squarefree energy] Energy loss = $E_{\text{total}} - E_{\text{sqfree}} = \log(abc/\text{rad}(abc)) = \sum (a_i - 1) \log p_i$ Energy loss = total repeated-factor energy = energy 'hidden' in squares/cubes. abc CONJECTURE = SQUAREFREE ENERGY DOMINATES TOTAL ENERGY: $H(e_c) = \log c < (1+\epsilon) \cdot E_{\text{sqfree}}(a,b,c) + O(1)$ INTUITION: When $a + b = c$, the additive relation FORCES certain prime factors to appear in $\text{rad}(abc)$ — the addition 'reveals' the primes. abc says: addition reveals enough primes that the squarefree energy of abc is almost as large as the energy of c alone. HOLOGRAM PICTURE: a, b, c are nodes on n_L . When $a+b=c$ (additive relation), the sub-hologram $\{e_a, e_b, e_c\}$ has a constraint that INCREASES its squarefree projection relative to its total tau-energy. abc = this squarefree projection is almost maximal.

4. Mason-Stothers Theorem: abc over Function Fields — PROVED in A_L

Over function fields (polynomials instead of integers), the abc conjecture is a THEOREM — the Mason-Stothers theorem (1983). In A_L, it follows immediately from the Riemann-Hurwitz formula for hologram coverings.

Theorem 4.1 (Mason-Stothers — proved in A_L).

Let k be a field of characteristic 0, and let a, b, c in $k[t]$ be coprime polynomials with $a + b = c$, not all constant. Then: $\max(\deg a, \deg b, \deg c) \leq \deg \text{rad}(abc) - 1$ (EQUALITY possible, unlike the integer case where $\epsilon > 0$ is needed) A_L PROOF: The polynomials a, b, c define a map $\phi: P^1 \rightarrow P^1$ via a/c (Belyi-type). In A_L: ϕ is a hologram covering $n_{\{P^1\}} \rightarrow n_{\{P^1\}}$ of degree $d = \max(\deg a, \deg c)$. The Riemann-Hurwitz formula for this covering: $2d - 2 = \sum \{P \text{ ramification}\} (e_P - 1) = (\text{total ramification of } \phi)$ The ramification occurs ONLY at zeros of a, b, c (because $a+b=c$) and at infinity. Counting: total ramification $\geq d - \deg \text{rad}(abc) + (\text{contribution at inf})$ Substituting into Riemann-Hurwitz: $2d - 2 \geq d - \deg \text{rad}(abc) \Rightarrow d \leq \deg \text{rad}(abc) - 1$ QED In A_L: this is the HOLOGRAM COVERING ENERGY INEQUALITY: $\tau\text{-energy}(\text{covering}) \leq \tau_{\text{rad}}(\text{base}) - 1$ The function-field abc = Riemann-Hurwitz for hologram coverings.

The Mason-Stothers theorem is EXACT (no epsilon) because over function fields, the arithmetic is GEOMETRIC — the Riemann-Hurwitz formula gives an exact count. Over number fields, the 'geometry' is replaced by analytic number theory, and the epsilon loss appears because we cannot count prime ramification as precisely as geometric ramification. This gap — geometry vs number theory — is what Mochizuki's IUT attempts to bridge.

5. What abc Implies: A Constellation of Theorems

The abc conjecture, if proved, would imply an extraordinary range of results. In A_L, each implication becomes an energy inequality.

Consequence	Classical statement	A_L formulation
Fermat's Last Theorem (for $n \gg 0$)	$x^n + y^n = z^n$ has no solutions for $n > 6$	High-power nodes have energy ratio > 1 — contradiction with abc

Mordell conjecture (Faltings 1983)	Curves of genus ≥ 2 have finitely many rational points	Rational points = tau-coincidences; abc bounds their number
Szpiro conjecture (for elliptic curves)	$\text{disc}(E) < C \cdot \text{rad}(\text{disc}(E))^6$ for all elliptic curves $E/\mathbb{Q}$	tau-energy of discriminant bounded by squarefree energy
Catalan / Pillai	$a^p - b^q = 1$ has finitely many solutions	Near-perfect-power nodes have limited energy slack
Wieferich primes rare	Almost no p with $2^{p-1} \equiv 1 \pmod{p^2}$	High-multiplicity nodes violate squarefree energy bound
Brocard's problem (dv290)	$n! + 1 = m^2$ only for $n=4,5,7$	Factorial energy exceeds square energy beyond $n=7$

Fermat's Last Theorem from abc.

abc implies FLT for all exponents $n > 6$ (unconditionally). In A_L : if $x^n + y^n = z^n$ with $\gcd(x,y,z)=1$, then $a = x^n$, $b = y^n$, $c = z^n$ satisfy $a + b = c$. The radical: $\text{rad}(abc) = \text{rad}(xyz) \leq xyz \leq z^3$. By abc: $z^n \leq K(\epsilon) \cdot (z^3)^{1+\epsilon}$. For large n : $n > 3(1+\epsilon)$, this is impossible for large z . In A_L : high-power nodes $e_{\{z^n\}}$ have energy $n \cdot \log z$, but squarefree energy $\text{rad}(z^n) = \log z$. The energy ratio $n/3 > 1$ for $n > 3$ — violating abc quality bound.

6. Mochizuki's IUT = Hologram Phase Transitions

Shinichi Mochizuki (2012) claimed a proof of the abc conjecture via his Inter-Universal Teichmüller Theory (IUT). The proof is extraordinarily complex and still under verification. In A_L , IUT has a natural interpretation via the hologram lifecycle (dv280).

The IUT architecture in A_L .

Mochizuki constructs: (i) Hodge theatres: isolated copies of the mathematical universe (ii) Log-theta links: connections between Hodge theatres (iii) The log-link: the logarithm map $\log: O^\times \rightarrow O$ (iv) The theta-link: a non-scheme-theoretic identification In A_L (via dv280, The River of Reality): Hodge theatre = A_L at inverse temperature β_i (isolated hologram) Log-theta link = Phase transition from β_i to β_j Log-link = $H_p = \log_p N$ (p -adic Hamiltonian, dv302) Theta-link = GT automorphism connecting two hologram copies The abc conjecture in IUT: The quantity $\log(\text{rad}(abc))$ is INVARIANT across log-theta links (it is preserved by the phase transition from one hologram to another). While $\log(c)$ can increase by at most epsilon under the link, the squarefree energy $\log(\text{rad})$ is conserved. abc = the squarefree energy is a hologram-lifecycle INVARIANT.

The indeterminacies.

IUT introduces three 'indeterminacies' (Ind1, Ind2, Ind3) that account for the freedom in identifying two different holograms. In A_L : the three indeterminacies correspond to three types of GT automorphisms that can change the identification between holograms. Mochizuki's key estimate: the three indeterminacies together contribute at most an epsilon loss — giving the abc inequality. In A_L : the three GT indeterminacies are three different types of $\text{Aut}_\tau(A_L)$ acting on the path-torsor of π_1^{mot} (dv304), each contributing a bounded energy perturbation.

The log-volume and tau.

The central invariant in IUT is the 'log-volume' of a Gaussian integral over the Galois action. In A_L : the log-volume IS the tau-energy of the corresponding sub-hologram measured at the infinite prime. The IUT estimate: $\log\text{-vol} \geq \log \text{rad}(abc) - \epsilon \cdot \log c$. In A_L : $\text{tau-energy}(\text{squarefree sub-hologram}) \geq \text{tau-energy}(c) - \epsilon$. This is the abc inequality.

7. abc as a Statement about Motivic Heights

The Weil height extends from rational numbers to motives. The abc conjecture is then a statement about motivic heights of the motive $M(a+b=c)$ — the motive encoding the additive relation.

Definition 7.1 (Motive of an abc-triple).

For $a + b = c$ with a, b, c coprime, define the abc-motive: $M(a, b, c) =$ motive of the Belyi map $\beta: P^1 \rightarrow P^1$ defined by $z \mapsto a^2z + b^2/c$ (mapping 0 to b/c , 1 to 1, ∞ to ∞). This is a mixed Tate motive in $MT(\mathbb{Z})$ (since it factors through $P^1 \setminus \{0, 1, \infty\}$). Its periods (dv303, dv304) are MZV evaluated at the abc-data. The motivic height of $M(a, b, c)$ in A_L : $h_{\text{mot}}(M(a, b, c)) = \tau(H \text{ restricted to } M(a, b, c)) = \tau_{\text{rad}}(e_a) + \tau_{\text{rad}}(e_b) + \tau_{\text{rad}}(e_c) = \log \text{rad}(abc)$ abc CONJECTURE = MOTIVIC HEIGHT INEQUALITY: $h(e_c) = \log c < (1+\epsilon) \cdot h_{\text{mot}}(M(a, b, c)) + O(1) = (1+\epsilon) \cdot \log \text{rad}(abc) + O(1)$ The total energy of c is bounded by the motivic height of the triple.

Theorem 7.2 (abc = Northcott property for abc-motives).

The Northcott property for heights states: for any bound B , there are only finitely many algebraic points of height $\leq B$. For abc-triples, the Northcott property applied to $M(a, b, c)$ gives: For fixed ϵ and K : $\#\{(a, b, c) : a+b=c, q(a, b, c) > 1+\epsilon\}$ is finite. This is EXACTLY the abc conjecture! In A_L : the abc conjecture = the Northcott property for the motivic height h_{mot} on the category of abc-motives $M(a, b, c)$. The finiteness of high-quality triples = the finiteness of abc-motives whose tau-energy greatly exceeds their motivic height. This is the motivic reformulation: abc = Northcott for abc-motives.

8. abc in the Grand Thread: Connection to RH and BSD

The abc conjecture is not isolated — it connects to RH and BSD through the common framework of A_L .

CONNECTION TO RH (via zeta zeros): Granville-Stark (2000): ABC $\Rightarrow$ no Siegel zeros for Dirichlet L-functions In A_L : abc $\Rightarrow H_{\Phi}$ -norm bounded $\Rightarrow B = \tau(H_{\Phi} H_{\Phi}^*) = 0 \Rightarrow$ RH The abc energy bound prevents 'rogue' zeros off the critical line. CONNECTION TO BSD (via elliptic curves): Szpiro conjecture (implied by abc): $\text{disc}(E) \ll \text{rad}(\text{disc}(E))^6$ In A_L : energy of discriminant bounded by squarefree energy Masser-Wustholz (1990): abc $\Rightarrow$ finite $\text{Sha}(E)$ for all $E/\mathbb{Q}$ In A_L : finite $\text{Sha} =$ finite $\dim \ker(T_{\{\Phi_E\}} - I)$ (dv216) CONNECTION TO MOTIVES (via periods): abc $\Rightarrow$ bounds on periods of mixed Tate motives In A_L : tau-energy of MZV bounded by rad-energy of arguments e.g., $|\zeta(n)| < \text{rad}(n)^{1+\epsilon}$? — new direction! THE THREE PILLARS: RH: zeros of zeta constrained by energy $B = 0$ BSD: rank of $E(\mathbb{Q})$ constrained by $\dim \ker(\text{hologram operator})$ abc: energy of sum constrained by squarefree energy ALL THREE = statements about tau-energy bounds in A_L .

9. The abc Architecture: What Remains

abc IN A_L — COMPLETE ARCHITECTURE STATEMENT: For coprime $a+b=c$: $H(e_c) < (1+\epsilon) \cdot \tau_{\text{rad}}(e_{\{abc\}}) + O(1)$ **PROVED CASES:** Function fields: EXACT equality by Riemann-Hurwitz [Mason-Stothers, §4] n -th power triples: abc quality $q = n/(3+\epsilon) < 1$ for $n < 3$ [dv305 §5] Specific families: Fermat curves, Catalan, Brocard (dv290) [dv305 §5] **ARCHITECTURE (gaps identified):** Main gap: extending Riemann-Hurwitz from function fields to number fields = extending hologram covering argument from geometric to arithmetic = what Mochizuki's IUT attempts (phase transitions between holograms) **MOCHIZUKI CONNECTION (dv280 + dv305):** Hodge theatres = holograms at different beta Log-theta links = phase transitions Log-volume = tau-energy at infinite prime abc = squarefree tau-energy is lifecycle-invariant **MOTIVIC REFORMULATION (dv303 + dv305):** abc = Northcott property for abc-motives $M(a,b,c)$ Quality $q > 1+\epsilon$ = motivic height $\gg$ total energy Finiteness of high-quality triples = Northcott for $MT(Z)$ **CONNECTION TO FULL CAMPAIGN:** abc $\Rightarrow$ no Siegel zeros $\Rightarrow$ GRH strengthened abc $\Rightarrow$ Szpiro $\Rightarrow$ finite $\text{Sha}(E) \Rightarrow$ BSD architecture abc + IUT = hologram lifecycle invariance abc = **ENERGY CONSERVATION FOR ARITHMETIC ADDITION** $H = \log N$: height of every number tau_rad: squarefree energy = motivic height abc: addition conserves motivic height (up to epsilon) **ONES IS FOREVER. WE DO. WE ARE. HU HU HU!!!**

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 $H(e_c) < (1+\epsilon) \cdot \tau_{\text{rad}}(e_{\{abc\}}) + O(1)$. abc = energy conservation under addition. Mason-Stothers proved. Mochizuki = hologram phase transitions. Addition vs multiplication. The fundamental tension of arithmetic. Chung ta la Hon Su. **ONES IS FOREVER. HU HU HU!!!**